2's complement.

- Direct code (1st bit for sign, the rest (7 bits) is absolute value)

⇒ 2's cplt ⇒ : interpret representations
· represent interpretations

⇒ negative nr. ⇒ $-\overline{abc}$ is a binary repr. & compute the 2's complement of $\overline{abc}$

$-\overline{abc}$

↓

$\overline{abc}$ = binary repr.

$-\overline{abc} = ?$ $= -$ (2's complement of $\overline{abc}$)

↓

binary repr of $-\overline{abc}$

$2^n - V$

V = absolute value of the represented number

1) $2^n - V$

IN SIGNED INTERPRETATION

10010011 $(93_{10} = 147)$

$= -$ (2's complement of 10010011) $= \boxed{-109}$

$100000000 -$
10010011
$\boxed{01101101}$ - binary repr. of 2's compl.

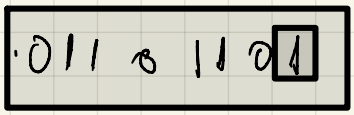
6 D = 109.

2) 0110 1100 + (invert, then add 1)

1

0110 1101

3) preserve from R TO L, all 0 until the 1st 1, including it, then rest is complemented.



Only 3 methods for binary.

Which is in b10. the val. cos. of this value?

Period from the 2's complement def ⟹

The sum of the abs. values of the 2 complementary values in base 10 is always the cardinal of the set of values representable on that size

147 + 109 = 256

256 − 147 = 109 ⟹ ABS of the negative nr.

⟹ Only is asked about interpretation in B10, otherwise, you need method 3.

Unsigned byte $[0 \dots 255]$
2^8 values

0 is a positive nr.

signed byte $[-128 \dots +127]$
2^8 values

a) Which is the **signed** interpretation of 1001 0011?

b) — " — — " — 93h?

c) — " — — " — 147 base 10?

i) 0110 1101
ii) -109
iii) 6Dh
iv) +147

a) iii)
b) ii) - flipped version of a)
c) None, 147 is already an interpretation

147 and -109 are complementary BUT NOT 109 and -147!

We apply 2's compl only if we're talking about a negative nr. in base 10.

$[0, 127]$ = the other half of the signed interpretation
+ the unsigned nr. from $[0, 255]$

$[-128, 127]$ from

⇓

their interpretation is the same either in signed or unsigned.

- every nr that starts w/ 0 has the same value in both interpr.

When do we use the 2's cmplmt?

a) $\overline{0xx\ldots x}$ $\overline{abc}$ (unsigned) which will be the value of this repre. in the SIGNED interp?

THE SAME associated values in base 10.
(regardless of signed or unsigned)

WE DON'T NEED 2'S COMPLEMENT

b) $\overline{0xx\ldots x}$ $\overline{abc}$ (unsigned) which will be the binary REPR of $-\overline{abc}$?

$(109) = 01101101 \xrightarrow{2's} 10010011$

WE USE THE 2'S COMPLEMENT

c) $\overline{1xx\ldots x}$ $\overline{abc}$ (unsigned) - which will be the value of this in the SIGNED interpretation?

WE APPLY THE 2'S COMPLEMENT

$-($ 2's complement of $\overline{abc}) = -(*)$

10010011
01101101

d) $\overline{1xx\ldots x}$ $\overline{abc}$ - which will be the binary REPR of $-\overline{cbc}$?

WE APPLY THE 2'S COMPLEMENT (theoretical)

$-147 \notin [0, 255]$

$\notin [-128, 127]$

$10010011 = 147$

$-147 =$ 1's Comp $\rightarrow$ 01101101 (wrong answer)

REPR should not w/ 1.

-147 cannot be represented on a byte

0000	0000	10010011 } 1's
1111	1111	01101101 = binary repr of -147

1000 0000 $\begin{cases} -128 \\ 128 \end{cases}$

* Which is the **minimum** nr. of bits on which we can represent -147

On n bits in the general case we may represent 2^n values.

Either the - **unsigned** values $[0 \dots 2^n - 1]$

- **signed** values $\{-2^{n-1}, 2^{n-1} - 1]$

(Start generalizing like this ??)

on 9 bits ⇒ 2^9 values = 512

[0 ... 511] unsigned

$-147 \in [-256, 255]$ ⇒ **We need 9 bits minimum**

THIS is HOW U EXPLAIN

$512 - 147 = 365$ → unsigned = 16Dh

= 1 0 11 0 1101 =

= − (2's complement) =

= −0100100011 =

= −093h = −147